

Slide 1 | Machine Learning for Satellite Link Fade Nowcasting

Target: 0:20 | Cumulative: 0:20

Good morning. Today I will present my experimental master's thesis, carried out at M.B.I. S.r.l. The work investigates how machine learning can use satellite signal measurements to support decisions about switching to a backup communication link during a fade.

Transition: I will begin with the application context and the operational problem behind the experiments.

Slide 2 | Industrial Context and Operational Problem

Target: 1:20 | Cumulative: 1:40

The application context is satellite communications. The study uses real fade and beacon recordings acquired at the Fucino ground station and examines a practical decision: when the satellite link degrades, is the fade likely to persist long enough to justify activating a backup link?

Switching too late can leave the link exposed to degradation. Reacting to every short fluctuation can instead cause unnecessary backup activation. The goal is therefore to anticipate persistence, rather than simply detect that a threshold has been crossed.

This is an experimental study of that decision, using historical data and an offline reference, not a deployed controller. The evaluation uses a ten-decibel threshold and a five-minute persistence horizon. At each decision time, the predictive models receive recent signal history, not the future observations used to assess their decisions.

Transition: This motivates a comparison of different ways to formulate the prediction problem.

Slide 3 | Research Objective and Contribution

Target: 1:00 | Cumulative: 2:40

The contribution is a decision-level comparison of four formulations. We can predict a future trajectory, estimate a duration, classify an event, or estimate the probability that the remaining event duration exceeds a given horizon.

Each formulation has its own natural training objective. However, all of them eventually need to support the same operational action. I therefore compare both their native predictive performance and the switch sequences obtained from their outputs. An important part of the framework is making the evaluation domain explicit: the tasks do not necessarily produce decisions at the same timestamps. Without accounting for that difference, their scores would not be directly comparable.

Transition: The comparison starts from a common collection of signal measurements.

Slide 4 | Data and Experimental Setting

Target: 1:10 | Cumulative: 3:50

The recordings mostly contain quiet, near-baseline conditions; fades are sparse. Uniformly sampling the full series would risk making training focus on reproducing these long plateaus rather than the changes relevant to switching. We therefore select samples around fade episodes.

Forecasting and current-level persistence use windows extending ninety minutes before and after candidate onset. These include nearby background and recovery, not only above-threshold samples. Long-fade and survival instead sample inside their task-specific events.

Inputs use signal only, normally thirty observations at thirty-second sampling. One complete source is held out; development splits are chronological and event-aware, with model selection on validation. The resulting evaluation is event-focused, not a test of continuous monitoring over all raw recordings.

Transition: The inputs are related, but the four targets answer different questions.

Slide 5 | Four Predictive Formulations

Target: 1:20 | Cumulative: 5:10

Autoregressive forecasting predicts the next ten observations, covering five minutes. Current-level persistence predicts the time until the signal falls below a level defined relative to its current value, with delta equal to half a decibel.

Long-fade detection asks whether the entire grouped event lasts at least five minutes. This is an event-classification question, not a prediction of how much time is left. Survival persistence addresses that remaining-time question directly by estimating a probability distribution.

These distinctions matter. A decline below the current level does not necessarily mean that a fade has ended. Similarly, knowing that an event is long does not tell us exactly when to turn the backup link off. The different targets therefore imply different decision rules.

Transition: To assess those rules, we need an explicit reference and a common evaluation procedure.

Slide 6 | Perfect Switch and Evaluation

Target: 0:55 | Cumulative: 6:05

Perfect Switch is an offline reference constructed using the true signal evolution. It identifies persistent threshold intervals and is not available to a real-time predictor. The models are evaluated according to how closely their decisions agree with this reference.

Precision tells us how much predicted switch activity agrees with the reference. Recall tells us how much reference activity is recovered. We also measure total active time and the number of separate switch intervals. These capture operational differences that a single F1 score can hide.

Transition: Before measuring these quantities, we apply the same post-processing to the model-derived switch decisions.

Slide 7 | Shared Switch Post-Processing

Target: 1:05 | Cumulative: 7:10

This post-processing acts on the binary switch decision, not on the measured signal. First, each task converts its prediction into an immediate switch request. The conversion differs between tasks, but the following steps are shared.

In this example, the initial request lasts only two samples. Minimum-duration processing extends that run to ten samples. It extends short activations rather than removing them.

The holding rule then checks the current observed signal. If the switch was active at the previous sample and the signal is still at or above ten decibels, it remains active even when the immediate request is zero. Once the minimum extension is complete, recovery below the threshold allows it to turn off. All subsequent switch results refer to this final sequence.

Transition: I will now explain how each predictive formulation produces the initial switch request.

Slide 8 | Autoregressive Forecasting

Target: 1:40 | Cumulative: 8:50

The forecasting models receive a past window and predict ten future signal values. I evaluate recurrent models, PatchTST, ten separate XGBoost regressors, and Chronos in zero-shot mode. The trained model families include raw and context-standardized variants. Standardization uses the past window, and forecasts are converted back to the original scale before evaluation.

The immediate switch rule requires every predicted point to remain at or above the threshold. This is evaluated at the current decision time, rather than by mixing predictions issued at different times.

The raw GRU sequence-to-vector model has the lowest test RMSE. However, raw PatchTST achieves the highest switch F1 on the native autoregressive grid. The trajectory is useful, but average forecast error does not specifically weight mistakes that change the switch decision. This formulation is effective but indirect: we predict an entire trajectory to answer a persistence question.

Transition: The next formulation removes the trajectory and predicts a duration directly.

Slide 9 | Current-Level Persistence

Target: 1:25 | Cumulative: 10:15

This target measures time until the signal falls below its current level minus half a decibel. That is not the same as fade recovery: the signal could drop and still remain above ten decibels.

XGBoost has a median absolute error of twenty-seven seconds but a mean error of almost ten minutes. Useful switching does not contradict these large errors. For example, predicting ten minutes instead of one hour is a very large duration error, but both exceed the five-minute decision threshold. This example illustrates the distinction; it is not a measured decomposition of our errors.

The current-signal gate and shared post-processing also contribute to the final policy. XGBoost's

switches are conservative on this holdout, but this does not establish reliable duration estimation. The tested shapelet variants never activate because their predictions fail the five-minute gate when the signal is above threshold.

Transition: Classification offers another shortcut, but it changes the meaning of the prediction.

Slide 10 | Long-Fade Detection

Target: 1:20 | Cumulative: 11:35

Long-fade detection predicts whether a grouped event is long. Nearby threshold crossings can belong to the same group, and all eligible timestamps within that event share its label. The dataset therefore focuses on event regions rather than continuous background monitoring.

XGBoost, the TCN, and the shapelet classifier obtain very high classification scores. These scores are conditional on this event-based dataset and should not be interpreted as near-perfect online fade detection.

When the probabilities become instantaneous switch decisions, the mismatch becomes clear. The TCN recovers every reference-positive sample on its native grid, but precision is only about fifty-four percent. A long event retains its positive label even near its end and across temporary recoveries in the group. Recognizing a long event therefore does not tell us how much of it remains. The formulation is useful for event characterization, but its total-duration label is poorly aligned with precise switch timing.

Transition: Survival modeling focuses instead on how much of the current fade remains.

Slide 11 | Survival Persistence

Target: 1:35 | Cumulative: 13:10

Survival modeling estimates a distribution over the remaining event duration. Of the four formulations, this directly addresses the question of whether the current fade will persist for another five minutes. If acquisition stops before recovery, we know only that the event continued for at least the observed period. These right-censored samples are retained.

XGBoost-AFT models the duration distribution through an accelerated failure-time formulation. The TCN predicts conditional event probabilities over time bins, from which we obtain a survival curve. Both use the same underlying continuous-time labels.

Direct target alignment does not guarantee the best empirical policy. Probabilities need calibration assessment, and the operating threshold must be chosen independently of the test set. Here 0.65 was explored on the test set, so the reported switches remain exploratory rather than confirmatory evidence of superiority.

Transition: To compare the tasks themselves, we must also align their decision timestamps.

Slide 12 | Cross-Task Switch Performance

Target: 1:20 | Cumulative: 14:30

The tasks make predictions on different domains. In the main comparison, I map their decisions onto a shared reference grid and fill missing native decisions with zero. Coverage therefore becomes part of the result, rather than being ignored.

Current-level XGBoost achieves the highest observed F1 among these representatives and has no false-positive switch samples on this grid, but misses some reference activity. Survival XGBoost-AFT recovers more of that activity, with some false positives. Raw PatchTST is the strongest autoregressive representative, while long-fade detection prioritizes recall.

Survival's low overall timestamp coverage does not imply equally low recall: its native domain concentrates on active fades. It still needs fallback behavior outside that domain. These results identify different trade-offs, rather than a universally superior controller.

Transition: Those trade-offs also affect how long the backup remains active and how often it is activated.

Slide 13 | Active Duration and Switching Events

Target: 1:40 | Cumulative: 16:10

On the common grid, Perfect Switch is active for about two hundred and thirty-five minutes. Current-level XGBoost stays below that duration, reflecting its conservative behavior. PatchTST is close in total duration, while survival XGBoost-AFT stays active somewhat longer. Long-fade TCN reaches more than four hundred minutes.

Duration alone is still insufficient. Two policies can be active for the same total time but divide it into very different numbers of intervals. Fragmentation matters because each activation represents another transition between links. A switch interval is a contiguous run of positive decisions, not necessarily a separate physical fade event.

For example, within the native autoregressive evaluation, PatchTST produces twenty switch intervals compared with eighteen for Perfect. That is a within-task diagnostic. Counts on different native grids must not be combined as if they shared the same reference domain.

Transition: A single event makes these timing and fragmentation differences easier to see.

Slide 14 | An Event-Level Comparison

Target: 1:20 | Cumulative: 17:30

This event contains a short early fade and a longer interval later on, followed by additional activity. Every row uses the same time axis and reference. The step curves show final binary decisions after shared post-processing: the lower level is zero and the upper level is one.

The early reference interval is recovered by survival XGBoost-AFT and the long-fade classifier, but not by current-level XGBoost or raw PatchTST. During the sustained central fade, the representative methods agree more closely. Differences become visible again around recovery and the later intervals.

This example illustrates why neither total duration nor a single aggregate score tells the whole story. It also shows that post-processing contributes substantially to the final policy. The event is an illustrative case, not independent evidence of general superiority.

Transition: These findings are useful, but the experimental scope sets limits on what we can conclude.

Slide 15 | Limitations and Next Steps

Target: 0:55 | Cumulative: 18:25

The results come from one external test source, so they do not establish robustness across all terminals or weather conditions. More independent holdouts are needed. The survival operating point also needs independent calibration on validation data.

Finally, this is an offline experimental framework, not a deployed controller. A streaming implementation must audit data availability during preprocessing, maintain switch state, measure latency, and define what happens when a task cannot produce a decision. Comparing against an operational engineering controller would then be an important additional step.

Transition: Within these limits, the experiments support the following conclusions.

Slide 16 | Conclusions

Target: 0:50 | Cumulative: 19:15

The four formulations have distinct strengths and limitations. Forecasting supports useful decisions, but average forecast error is not a switch-specific objective. Current-level XGBoost can switch conservatively despite large duration errors; its target is only a proxy for fade persistence. Long-fade classification recognizes long events but does not estimate how much time remains. Survival addresses that residual-duration question directly, although calibration and independent evaluation are still needed.

We therefore have different operating trade-offs, not a universal winner. The central conclusion is to evaluate target alignment, prediction quality and the final switch policy together.

Slide 17 | Thank You / Questions

Target: 0:05 | Cumulative: 19:20

Thank you for your attention. I am happy to take your questions.

End of the main talk. The following pages are optional Q&A notes.

Questions: backup-slide guide

Not part of the timed speech. Backup slides follow the closing screen in the deck.

B1 | How does each model activate a switch?

Four explicit raw-decision rules, followed by shared post-processing.

B2 | What exactly is Perfect? Does minimum-island remove short events?

Eleven reference observations span 300 s; short decision islands are extended, then held above threshold.

B3 | Which architectures and input variants were evaluated?

Full family matrix, including Chronos L120 and selected survival representations.

B4 | How extensive was model selection?

Archived trial counts and validation-only selection criteria.

B5 | How are inputs normalized and gaps handled?

Context-only normalization, train-fitted scalar scaling, and offline interpolation availability caveat.

Questions: backup-slide guide (continued)

B6 | Why are event definitions and censored samples different?

Current-level recovery, grouped events and survival recovery are distinct targets; acquisition gaps bound observation.

B7 | How can large duration error coexist with useful switches?

Illustrative duration-threshold counterexamples; not empirical samples or a claim of perfect target alignment.

B8 | Is cross-task comparison fair?

Native, common-reference and strict-intersection domains answer different questions.

B9 | How can duration labels produce a survival probability?

AFT time distribution versus TCN conditional hazards; both account for censoring.

B10 | Is the survival ranking statistically meaningful?

Event-bootstrap C-index interval includes zero; probabilistic accuracy and switch F1 are different.

Questions: backup-slide guide (continued)

B11 | How does PatchTST turn a signal into forecasts?

Thesis architecture figure, credited to Nie et al. (ICLR 2023). Selected raw model: 30 samples, patch length 10, stride 2, 11 tokens, width 128, four heads, three layers, ten outputs. The original figure includes instance normalization; our model instead receives raw or externally context-standardized input.

B12 | Exactly what enters each autoregressive model?

GRU and PatchTST use a 30-by-1 raw or context-standard sequence; XGBoost receives the same 30 values as tabular lags. Chronos takes 30 or 120 raw values and performs internal scaling/tokenization. No scalar features are appended. Each model predicts ten future signal values.

B13 | Exactly what enters the persistence models?

Current-level uses relative-to-current lags plus 12 scalars; long-fade uses relative-to-threshold lags plus 14. Selected survival AFT uses 30 raw lags only; selected survival TCN uses relative-to-current history plus 18 scalars. Neural models embed scalars separately and concatenate them with the sequence encoding; trees concatenate scalar values directly with lags.

B14 | What are the scalar features?

Ten common summaries: full-window mean, standard deviation, min, max, range; changes over one, two and four intervals; last-five mean and standard deviation. Current-level adds current signal and current-minus-delta. Long-fade adds current signal, threshold distance, elapsed event time and sample count. Survival adds four online threshold-state summaries. Scaling uses training data during selection; no future target enters an input.

Questions: additional details

Slide 4: Methodological clarification for questions

plateau dominance is the sampling rationale, not a demonstrated full-series versus event-window ablation. Cropping selects eligible prediction timestamps; it does not join distant observations or allow contexts to cross acquisition gaps. Current-level duration labels are searched on the full continuous signal, not truncated at the candidate-window boundary. Timestamp validation and conservative small-gap handling still apply (B5/B6). Chronos is zero-shot; the survival switch threshold has the explicitly stated exploratory caveat.

Slide 9: Additional native-test values for Q&A

XGBoost switch precision 1.000 and recall 0.822. These remain off the main slide.

Slide 10: Additional native-test values for Q&A

TCN switch F1 0.702 and active duration 445 min. These are native-grid values, not the common-grid results shown later.

Slide 11: optional numerical comparison

If useful in the spoken explanation: the TCN has the lower integrated Brier score, while XGBoost-AFT produces the better switch sequence at the evaluated operating point. The numbers are in B10: IBS 0.103 versus 0.117, and switch F1 0.784 versus 0.863, respectively. Keep this numerical detour for questions if time is tight.